CarePath AI

An accessible, privacy-conscious healthcare information and navigation assistant

1. Project overview

CarePath AI is a proposed research prototype that helps people understand public NHS information and identify appropriate human support channels. It does not diagnose conditions, determine clinical urgency using an AI model, interpret individual medical records as clinical advice, or book appointments. Its core promise is simple: understandable information, traceable sources and a clear route to a person when help is needed.

2. Who it helps and why

User group	Problem today	Proposed benefit
People with limited health literacy	Health terminology and long webpages are hard to follow	Plain-language summaries linked to original guidance
People facing language barriers	Information may be difficult to find or understand in English	A carefully scoped second-language option, with safety checks
Mobile or low-bandwidth users	Heavy websites and complex journeys impede access	Lightweight, responsive, text-first interface
Carers and community support workers	Need to help people navigate trustworthy information	Repeatable explanations and clear handoff instructions

3. Challenge fit

The hackathon poster names bias and fairness, explainability, privacy and security, and sustainability and access. CarePath connects each theme to an observable user-facing behaviour rather than a claim made only in a presentation.

Hackathon theme	Observable feature
Bias and fairness	Compare equivalent synthetic questions phrased differently; report inconsistent answers
Explainability	Show the exact retrieved NHS source, limitations and reason for a human handoff
Privacy and security	Redact identifiers before model input; minimise logging; limit tools to approved guidance
Sustainability and access	Use a small local model, cache public material, and offer a text-first accessible UI

4. Scope and success definition

The minimum viable product (MVP) completes one reliable journey: user question → privacy check → deterministic safety gate → trusted retrieval → bounded answer with citations → understandable next step. A successful demo includes a normal question, an emergency escalation and a redaction example. No real patient data is required or appropriate.

5. Proposed architecture

A small Python backend orchestrates a controlled pipeline. Streamlit is the quickest interface option; FastAPI and a pre-existing frontend are also suitable if already working. The model is a bounded explanation component, not a clinical decision-maker.

Stage	Function and control
1. Accessible intake	Simple mobile-friendly text form; plain language; optional language choice; keyboard and screen-reader labels
2. Privacy layer	Redact detected names, contact details and other identifiers; avoid storing raw questions by default
3. Safety gate	Deterministic rules detect a small, documented set of emergency cues and show verified escalation instructions; unclear cases are not labelled safe
4. Retrieval layer	Search a curated, versioned set of official NHS pages; retain URLs and evidence excerpts
5. Bounded response	Small local model rewrites supported content in simple language; no sources means abstain or human handoff
6. Output and evidence	Visible source links, limitations, next steps, and a human-support option
7. Evaluation and monitoring	Synthetic test cases; accuracy of citation links; safety, privacy, fairness and resource metrics

6. How sustainability and access are built in

Access operates at both ends of the journey: low-friction intake and readable output. The MVP prioritises a responsive, text-first design, keyboard use, clear contrast and plain language. One additional language is a stretch goal only if it can be tested safely; voice and messaging platforms are outside scope.

Sustainability operates inside the system: use a right-sized local model, retrieve rather than fine-tune, cache non-sensitive public guidance, and avoid repeated model calls when a deterministic answer is sufficient. Record model size, median response time and memory use; estimate energy use only when a credible measurement method is available. Local inference does not automatically prove a lower carbon footprint.

7. Boundaries, risks and mitigations

Risk	MVP mitigation
Unsafe clinical-sounding answer	Model cannot override safety gate; fixed escalation messages and abstention
Fabricated or outdated sources	Curated NHS corpus and citation verification; state retrieval date
Privacy leakage	Synthetic data only; redact before inference; log metadata rather than question bodies
Unequal language performance	Same-intent paraphrase tests; explicitly label untested languages
Prompt injection in source text	Treat retrieval as untrusted data; no ability to modify records or perform external actions

8. User journey

The product should be demonstrated as a user experience, not merely a technical pipeline. Each screen answers the user's next practical question and avoids dead ends.

Moment	What the person experiences	How the design takes care of it
Arrive	"Can this help me?"	One-sentence purpose, limitations, clear start control and privacy notice
Ask	"How do I describe my problem?"	Open text, helpful examples, simple wording and no compulsory account
Protect	"What happens to my information?"	Identifiers redacted; short visible privacy explanation; no default storage
Get support	"Is this urgent?"	Emergency cues trigger a direct human/999 handoff before AI generation
Understand	"Where did this answer come from?"	Short response, working NHS source link and uncertainty wording
Act	"What should I do next?"	Relevant NHS or human-support pathway, without pretending to book or diagnose
Recover	"What if it cannot answer?"	Explicit abstention, source unavailable message, link to human help

9. Three demo stories

- Normal navigation: a user asks what a term in public NHS guidance means. The assistant gives a short, sourced explanation and a next step.
- Safety: a synthetic user message includes a documented emergency warning sign. The deterministic safety gate bypasses generative advice and displays the appropriate escalation instruction.
- Privacy: a synthetic question contains a name and telephone number. The UI demonstrates the redacted text that reaches the local model.

10. Evaluation that judges can inspect

Target around 20 synthetic cases spanning the three journeys, missing evidence, paraphrases, optional language variants and prompt injection attempts. Record expected outcome, actual outcome, pass/fail, source validity and any limitation. Report the results honestly; these tests do not establish clinical validity or overall fairness.

11. Weekend execution priorities

When	Output and stop condition
Saturday morning	Complete end-to-end normal journey with verifiable NHS evidence
Saturday afternoon	Implement safety, privacy and three repeatable demonstrations
Saturday evening	Feature freeze; run automated tests and document limitations
Sunday before 10:30 AM	Finish user-journey walkthrough, pitch, repository and backup demo
Sunday 10:30–11 AM	Submission checks and contingency buffer; no new features

12. Responsible claim and next steps

CarePath AI is an unvalidated hackathon prototype for healthcare information navigation, not a medical device or substitute for professional care. A future phase could involve accessibility research with users, language-specific evaluation, independent clinical review, and more rigorous energy measurement. These are not MVP claims.

Basis: UWE hackathon poster supplied by the team and the three-page "Responsible AI Hackathon UWE: Healthcare, Agentic AI, Access and Sustainability" approach guide. The name CarePath AI, user journey, engineering choices and delivery plan are proposed design decisions, not confirmed event requirements.